

Discovering Effective Variables from Interaction

A Research Position on Relational Latent World Learning

1 What problem are we solving, and why is it worth solving?

Consider a robot pushing a cup from one side of a table to the other. The cup moves with the hand while contact is maintained; the tabletop supports the cup and prevents motion through the surface; changing the cup, the surface, or the push strength changes the exact trajectory. A useful agent can learn from these regularities that the cup is movable, contact transmits motion, and the table supports and constrains objects. It does not first need to recover the full physical ontology of force, friction, and gravity. For prediction and action, these recurring effect patterns can serve as *effective variables*: compact variables that describe what can change and how that change behaves within a domain.

This capability is still limited. Predictive models can compress observations and forecast future states, skill models can summarise behaviour, and graph networks can model interactions. Yet the learned variables may change meaning when objects, strengths, or combinations change. The same observed prediction can also be produced by several incompatible internal decompositions. This is a general identifiability problem: the data and objective may be insufficient to decide which latent explanation should count as the same variable. Results on unsupervised disentanglement make this limitation precise—without additional assumptions or evidence, the intended latent factors are not identifiable from observations alone .

Leaving this unresolved has practical and scientific consequences. A robot may succeed in familiar scenes while failing to reuse what it learned on a new object or material. It cannot reliably explain whether a failure came from a previously known mechanism, a new mechanism, or a changed relation. It also lacks a principled way to choose the next interaction that would distinguish these explanations. Solving the problem would enable reusable prediction, composition of known effects, diagnosis of failure, and active collection of missing structural evidence across manipulation, tool use, and other interactive domains. These capabilities go beyond improving one benchmark score.

Can an agent discover variables that are useful for prediction and action from continuous interaction, learn their reusable effect laws and stable relations, and improve both their generalisation and identifiability through further interaction?

The first investigation is deliberately narrower. It asks whether a short observed change can be explained by stable, reusable mechanism functions. The E0 experiments set all mechanism relations to zero and study only the transition from TPs to intrinsic mechanism structure. Directed relations are introduced only in the next stage.

2 How have people approached this problem, and what remains difficult?

Predictive and object-centric world models. Object-centric models separate a scene into persistent entities and can improve compositional prediction . Predictive world models similarly learn compact state transitions that support planning. Their training objective, however, usually

constrains predicted observations rather than the identity of the hidden explanatory units. Several latent decompositions can therefore make similar predictions: one mechanism can be split by context, several mechanisms can be mixed, or their roles can be exchanged. Prediction is essential evidence, but it does not by itself make the variables identifiable.

Skills, options, and latent actions. Temporal abstractions compress action sequences into reusable control units . Latent-action models extend this idea to data without action annotations. Their variables primarily describe *how the agent acts*. Action meaning depends on the body and the current state: the same command can produce different changes in different contexts, while different commands can produce the same change. Recent results show that nuisance variation can dominate latent-action learning without suitable supervision . An action representation therefore cannot be assumed to represent a domain-level mechanism. A TP in this paper instead represents *what actually changed*.

Disentangled and causal representation learning. Identifiability can be recovered by adding structural assumptions, auxiliary variables, environment changes, or interventions . Temporal causal representation methods such as CITRIS and BISCUIT use intervention information to constrain latent causal variables ; later work extends identifiability results to multi-node interventions . These results establish that interventions can supply missing information. Their conditions do not directly solve our setting: a robot action may activate several mechanisms, the action label is not a mechanism label, and one observed TP may be a composition of multiple effects.

Graph networks and relational inference. Graph networks provide an appropriate inductive bias for interactions among entities or variables . Neural Relational Inference can learn a latent interaction graph from trajectories . But a predictive edge need not have a stable causal meaning, especially when node identities are themselves unknown. Co-occurrence cannot determine direction, and a flexible relation module can absorb effects that should belong to the nodes. Relations must therefore be supported by directional interventions and learned jointly, but cautiously, with variable identity.

The unresolved obstacle is consequently more specific than inaccurate prediction. We lack a method that can form mechanism variables from interaction without structural labels, define them by reusable effect laws rather than sample similarity, and use interventions, held-out contexts, and relational evidence to test whether those variables are functionally identifiable. The proposed study turns that gap into explicit formation, comparison, and falsification tests.

3 What is our central idea, and why should it work?

Central conjecture. In a bounded domain and time scale, suppose interaction-induced changes are generated by a small set of reusable mechanisms and stable directed relations. If experience is organised successively into persistent state, comparable TP response families, mechanisms with shared realisation coordinates, and relations among mechanisms, and if these structures are repeatedly tested by held-out prediction and new interventions, then a compact relational world should be recoverable up to functional equivalence.

Functional equivalence is the relevant target. Learned mechanism numbers and internal coordinate systems may differ between runs, but mechanisms that represent the same variable must produce the same effects under matched states and interventions and must occupy the same relational role. The proposal does not require the learned variable to reveal the deepest physical essence of the world. In the cup example, “contact transfers motion” can be an effective mechanism even if the agent never names force or friction.

The conjecture combines five structural priors. First, useful domain behaviour is compressible at the chosen scale. Second, interventions and responses obey regularities. Third, a TP describes a locally coherent change regime. Fourth, mechanisms recur across contexts and combine locally. Fifth, some mechanisms have stable directed effects on others. A single push gives weak identity evidence, because its direction and magnitude are entangled with the current scene. Repeating a comparable local intervention at several strengths reveals a response family. The values may differ, but the way the response varies should be shared if the underlying mechanism is the same.

The learning process is also motivated by development and Bayesian accounts of learning. Infants use action experience to interpret subsequent actions, and children can use patterns of intervention and outcome to learn causal structure . Hierarchical Bayesian accounts describe conceptual development as building and revising structured models of the world . We use this as a design analogy: candidate mechanisms and relations are hypotheses, and new interactions change their empirical support. We call the process *Bayesian-style hypothesis updating*; the present implementation does not claim to perform exact Bayesian inference.

The idea is independent of a particular neural architecture. Transformers, graph networks, or other function approximators may implement the required maps. What must transfer is the evidence structure: comparable interventions, shared response coordinates, held-out functional tests, directional relation tests, and an action loop that seeks observations able to distinguish competing explanations.

If the conjecture is correct	Conditions that can make it fail
Mechanisms retain their effects on unseen states, strengths, and combinations.	The state representation omits information needed to predict effects.
Independent runs recover the same functions up to permutation and coordinate change.	TP boundaries or a shared realisation coordinate do not exist at the selected scale.
Redundant mechanisms can be removed without harming held-out prediction.	Optimisation or limited capacity mixes mechanisms or splits one by context.
Removing or reversing a true relation damages prediction in the appropriate conditions.	Available interventions cannot distinguish node effects from relational modulation.
Informative interactions reduce structural ambiguity faster than random interaction.	The TP prior cannot realise the required diagnostic changes through available actions.
Task knowledge transfers through a shared domain world.	Tasks do not share the same domain structure or cannot be expressed in TP space.

The final two claims are downstream scope rather than part of E0. They assume that tasks share the same domain world, that a task constraint can be extracted from multiple demonstrations, and that task progress can be described through transformations. The current experiments test only the mechanism-formation part of the central conjecture.

4 How do we turn the idea into a precise model?

Figure ?? summarises the full proposed architecture. Solid components define the present model; dashed components are required by the theory but have not been tested by E0.

Relational Latent World Learning: Overall Architecture

From interaction-grounded local change to reusable mechanisms, relational structure, active evidence collection, and task execution

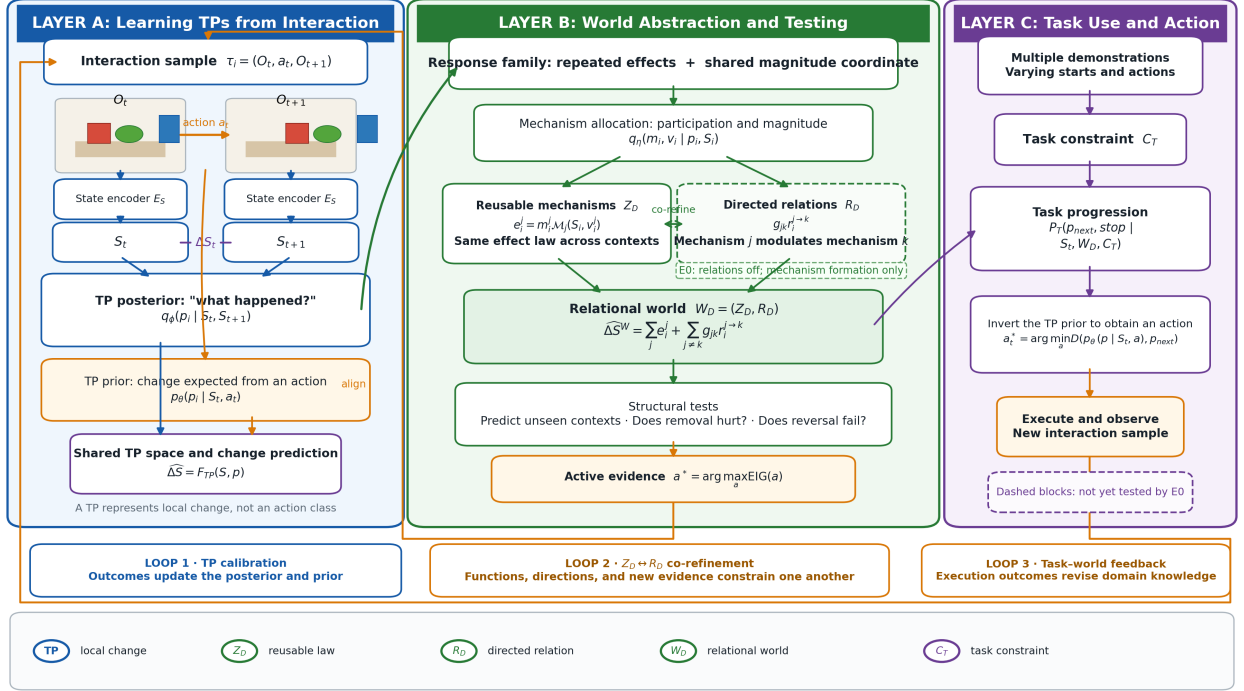


Figure 1: The proposed hierarchy and its three feedback loops. The TP posterior describes observed change from pre- and post-state; action enters the prospective TP prior. Mechanisms and relations form the domain world. Task execution and relation-guided exploration are future stages beyond E0.

4.1 Inputs, state, and transformation primitives

The input is a set of interaction sequences,

$$\mathcal{D}_{\text{world}} = \{(O_0, a_0, O_1, \dots, a_{T-1}, O_T)\},$$

where O_t is an observation and a_t is an executed action. A state encoder produces a persistent state S_t that retains information needed to measure and predict change. A TP p_i describes the local change between the pre-state and post-state.

Two complementary distributions prevent the TP from becoming merely an action code:

$$q_\phi(p_i | S_i^{\text{pre}}, S_i^{\text{post}}), \quad p_\theta(p_i | S_i^{\text{pre}}, a_i).$$

The posterior answers “what happened?” from the observed state change. The prior answers “what change is likely if this action is executed?” Action is therefore used to predict a TP, but it is excluded from the definition of an observed TP. Training aligns the two distributions and uses a TP decoder F_{TP} to predict the resulting state change.

Comparable repetitions of one local effect form a response family,

$$\mathcal{F}_r = \{(S_{r\ell}^{\text{pre}}, \alpha_{r\ell}, \Delta S_{r\ell})\}_{\ell=1}^{n_r},$$

where $\alpha_{r\ell}$ records the relative strength of the executed intervention. In the first mechanism experiment, a learned mechanism uses a shared affine coordinate

$$v_{r\ell}^j = a_j \alpha_{r\ell} + b_j.$$

The absolute sign and origin may vary, but the ordering supplied by the physical intervention is shared across the response family.

4.2 Mechanisms, relations, and identity

The output of world learning is

$$W_D = (Z_D, R_D),$$

where $Z_D = \{Z_D^1, \dots, Z_D^{M_{\max}}\}$ is an overcomplete set of candidate mechanisms and R_D contains candidate directed relations. For experience i , mechanism j contributes

$$e_i^j = m_i^j \mathcal{M}_j(S_i^{\text{pre}}, v_i^j).$$

Here m_i^j says whether the mechanism participates, v_i^j describes its current realisation, and $\mathcal{M}_j$ is the reusable response law. Several mechanisms may participate in the same TP, so the mapping between TPs and mechanisms is many-to-many rather than a clustering problem. Before relations are introduced, the predicted change is

$$\widehat{\Delta S}_i^Z = \sum_j e_i^j.$$

A directed relation $j \rightarrow k$ is not an additional independent effect. It represents how mechanism j modifies the effect expressed by mechanism k . With a global edge gate g_{jk} and an instance-specific relation residual $r_i^{j \rightarrow k}$, the full world predicts

$$\widehat{\Delta S}_i^W = \sum_j e_i^j + \sum_{j \neq k} g_{jk} r_i^{j \rightarrow k}.$$

Direction cannot be inferred from co-occurrence. Evidence for $j \rightarrow k$ requires matched experiences in which the state and target mechanism are approximately held fixed while the proposed source mechanism changes. Removing the edge or reversing its direction must then reduce held-out explanatory power.

Mechanism identity is consequently operational rather than nominal:

$$\text{Identity}(Z_D^j) = (\mathcal{M}_j, \Sigma_j^{\text{int}}, \mathcal{R}_j).$$

$\mathcal{M}_j$ is what the mechanism does, Σ_j^{int} records how interventions engage it, and $\mathcal{R}_j$ is its incoming and outgoing relational signature. E0 observes only the first two parts; the next experiment asks whether the third resolves remaining ambiguity.

4.3 Learning objective and structural tests

The core objective for learning the domain world is

$$L_{W_D} = L_{\text{world}} + \lambda_R L_{\text{edge}}^{\text{support}} + \lambda_P L_{\text{participation}} + \lambda_B L_{\text{base}} + \lambda_T L_{\text{TP-rec}}.$$

Its terms have direct meanings:

- L_{world} makes mechanisms and relations jointly predict the observed change.
- $L_{\text{edge}}^{\text{support}}$ removes unnecessary relations only when the data were sufficient to test them.
- $L_{\text{participation}}$ discourages using too many mechanisms in one experience.
- L_{base} protects the intrinsic effects learned from approximately isolated mechanisms.
- $L_{\text{TP-rec}}$ weakly checks that the mechanism decomposition preserves the TP information.

A low value of this loss is necessary but insufficient. Variable identity is evaluated separately by prediction on unseen states, strengths, and mechanism combinations; by the increase in error after a mechanism or relation is removed; by relation-direction controls; and by recovery consistency over independent training runs. These tests compare functions rather than learned indices.

4.4 Closing the loop and defining the present boundary

When several structures fit existing data, the world model should select an executable interaction with high expected information gain,

$$a^* = \arg \max_a \text{EIG}(a; W_D),$$

subject to the competence of the TP prior. The outcome becomes a new sample and updates the TP, mechanisms, and relations. This is a planned active-evidence loop, not a result established by E0.

For downstream tasks, demonstrations of the same task are grouped to infer a task constraint C_T . A task-transition model proposes the next TP and a stopping decision,

$$P_T(p_{\text{next}, \text{stop}} \mid S_t, W_D, C_T).$$

An executable action is obtained by inverting the learned TP prior,

$$a_t^* = \arg \min_a D(p_\theta(p \mid S_t, a), p_{\text{next}}).$$

The task branch is included to show why effective variables matter, but it is not evaluated in the current paper.

E0 sets $R_D = 0$. It preserves multiple-mechanism composition, state dependence, intervention strength, an overcomplete mechanism bank, and held-out function tests. It omits relational modulation, visual-perception error, real-robot execution error, active data collection, and task learning. E0 can therefore test intrinsic mechanism formation, but it cannot establish recovery of the complete relational world.

5 What does the first experiment reveal?

5.1 The smallest controlled test

The smallest experiment tests the first distinctive implication of the conjecture: a mechanism should be learned from an ordered response law rather than unrelated per-observation values. The synthetic world contains three reusable mechanisms. The same mechanism produces different changes across states and intervention strengths. Data, response-family membership, prediction architecture, and training budget are held fixed. The baseline assigns each observation a free realisation value; the proposed condition makes all observations in a response family share an ordered realisation coordinate. Two negative controls either shuffle the order within each family or reverse it randomly.

This corresponds to a robot repeating the same local intervention at several strengths and comparing the resulting changes. The result supports the explanation if the shared coordinate improves prediction at unseen strengths and the improvement disappears when its physical ordering is destroyed. It challenges the explanation if free values perform equally well or if shuffling the coordinate has no effect.

The primary prediction measure is the mean prediction deviation divided by the typical magnitude of the true change. Thus an error of 0.02 means that the average deviation is about two percent of a typical effect. Functional recovery asks whether each learned law matches the same generating law on unseen states and strengths. Count recovery asks whether the model can begin with five possible laws and safely retain the three that are needed. Complete recovery requires held-out function, mechanism participation, count, and full-TP prediction tests to pass together.

5.2 Results and variation across runs

Figure ?? presents the evidence chain in reader-facing terms. In an initial single-run comparison, alternating the mechanism allocation and function updates reduced relative prediction error from

57.25% to 36.33%, but neither it nor joint training recovered the correct structure. All five available laws remained involved. This shows that improving prediction does not by itself resolve latent decomposition.

With fixed response families, changing from a free value for every observation to a shared intervention coordinate reduced unseen-strength error from 18.31% to 1.86%. The three learned response functions explained essentially all of the generating-function variation. When the intervention order was shuffled, error rose to 125.3%; a random reversal produced 120.8%. This was a single development run, so it establishes a strong controlled contrast rather than cross-run stability.

The subsequent stability test fixed the synthetic world, data, and reliable response families and varied model and optimisation initialisation ten times. All ten runs formed functions covering the three true effects. Nine also removed the two redundant slots automatically and passed the complete TP tests. Across these runs, unseen-strength error ranged from 1.39% to 2.43%; full-TP error ranged from 2.19% to 3.32% across familiar and new mechanism combinations; participation decisions were 99.68%–99.87% correct. This tests the later function-formation and pruning stages under reliable upstream evidence, not complete discovery.

When response-family membership, mechanism functions, and realisation coordinates all had to be learned, the best current method recovered the complete structure in two of three independent runs. It assigned each response family to one current explanation and periodically tested whether every learned law remained necessary. Methods that trained two or three explanations from every family recovered the structure in zero of three runs. The two successful runs had held-out errors of 3.03% and 1.69%; the failed run had 4.97% error but split one true effect by context. The small difference in prediction error again shows why structure must be evaluated separately. Only model and optimisation randomness changed across these runs; the evidence does not yet cover different worlds, datasets, or real robots.

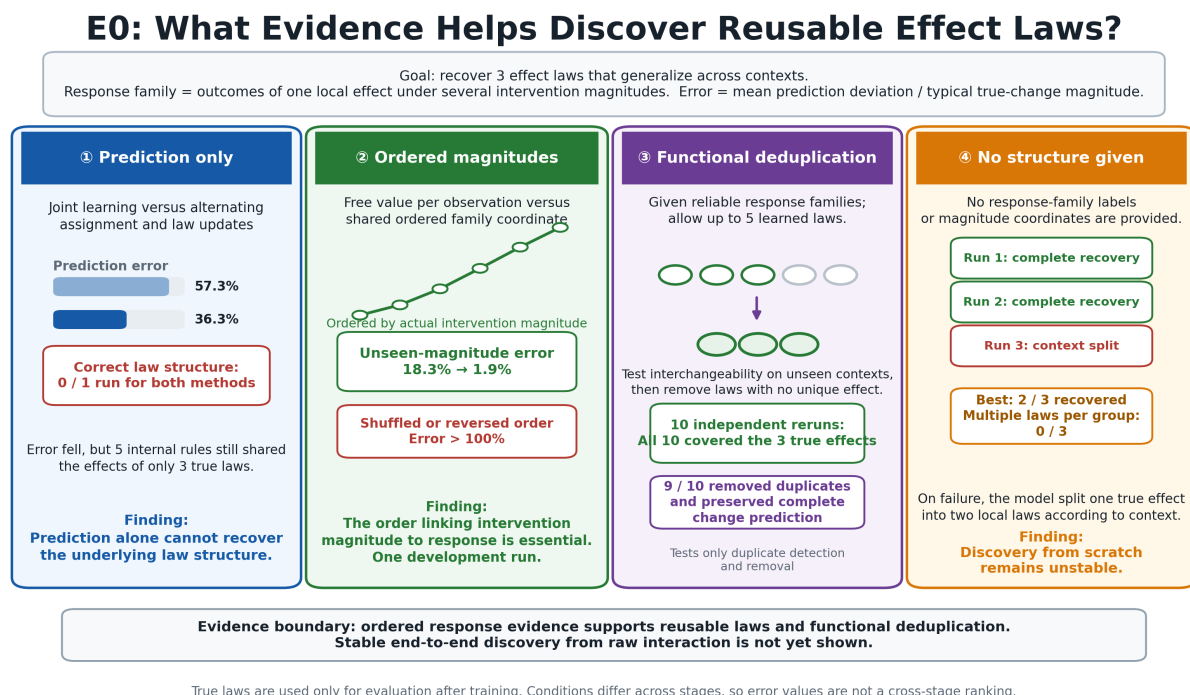


Figure 2: The E0 evidence chain. Each panel answers a different question and states its experimental boundary. Generating-mechanism information was used only after training for evaluation.

5.3 Concrete success and failure cases

The two cases in Figure ?? come from the same unsuccessful independent run. They were selected from the principal merge pairs proposed by the model’s own functional tests, before using true mechanism labels for interpretation. One case yields a controlled explanation; the other remains inconclusive.

In the successful case, two learned laws covered different state regions of one true effect. The hypothesis was that the retained law lacked enough mechanism-specific expressive capacity. Adding a small residual branch to that law, while freezing the shared state-processing trunk, reduced error on the newly absorbed region from 12.35% to 7.19%. Error on the law’s original region also improved by 1.76 percentage points. Giving the same additional capacity to a deliberately wrong law produced 55.73% error on the absorbed region and damaged its old predictions by 16.41 percentage points. The result supports a capacity-induced split for this pair: extra parameters alone were insufficient; the functionally compatible law had to receive them.

The unresolved case involved another pair in which one law appeared locally useful while the other was stronger across contexts. If the pair represented one function in different coordinates, a coordinate bridge should align them. A linear bridge explained only 39%–43% of their variation, and a state-dependent bridge explained 50%–57%. Learning a fresh union function, retaining two coordinates, and allowing protected changes to the shared representation all failed. Dividing the states into four data-derived regions did not reveal complementary specialisation: the more complete law performed better in every region. These observations make the weaker law look like a local identity-formation branch, but they do not prove that it is erroneous. E0 contains no relational evidence that could show whether the two laws occupy different roles.

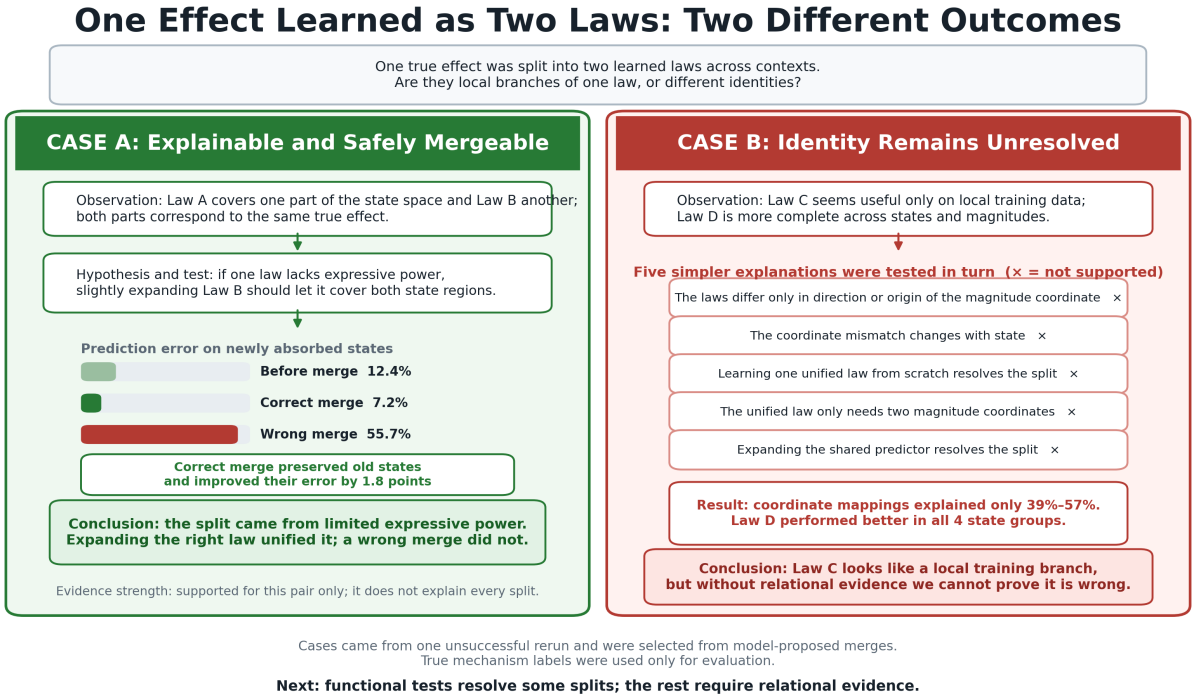


Figure 3: Two forms of context-dependent mechanism splitting. Functional evidence safely resolves the first pair. Five simpler intrinsic explanations fail for the second pair, leaving a mechanism-identity question for the relational experiment.

Taken together, the cases show why neither an illustrative success nor an aggregate error is enough. A proposed merge must preserve the old function, improve the newly covered region, fail for an incompatible comparison, and remain stable on held-out contexts. Even these tests may leave identity unresolved when only intrinsic effects are available.

6 What have we learned, and what comes next?

The present evidence supports three claims. First, the compositional mechanism representation can express and recombine the generating effects when adequate structural evidence is available. Second, mechanism identity is better supported by an ordered family of intervention responses than by a single observed effect. Third, once complete functions have formed, functional interchangeability across states, strengths, and combinations can often identify and remove duplicate laws.

The evidence does not establish stable discovery from raw interaction. It does not show robustness across different generated worlds, observation models, or real robots. It also says nothing yet about learning directed relations or the complete identity $(\mathcal{M}_j, \Sigma_j^{\text{int}}, \mathcal{R}_j)$. The main update to the original learning hypothesis is therefore procedural: sparse TP reconstruction is insufficient. Mechanisms should be formed from response families and shared coordinates, tested on unseen conditions, and pruned only after functional redundancy is demonstrated. Ambiguities that survive intrinsic tests should then be examined through relations rather than forced into a premature merge.

The key remaining uncertainty is whether relational evidence can improve mechanism identifiability without simply adding another flexible predictor. The bounded next stage, E1, will introduce a small synthetic world with known directed modulation. Some mechanisms will have deliberately similar intrinsic effects but different relations to other mechanisms. Under matched data, capacity, and initialisation distributions, E1 will compare:

1. an intrinsic-only mechanism learner;
2. a learner with relation modules but no direction-specific evidence test; and
3. the proposed learner with matched interventions, edge-removal and edge-reversal controls, and $Z_D \leftrightarrow R_D$ co-refinement.

The evaluation will measure held-out prediction across states and strengths, mechanism-function and mechanism-count recovery, directed-edge recovery, and consistency across independent runs. A positive result requires relational evidence to reduce context splitting, recover the correct direction more consistently than the baselines, and preserve already learned intrinsic effects. If the forward and reverse relations remain interchangeable, or if relations improve prediction while mechanism identity remains run-dependent, the conclusion will be that the available intervention coverage or relation definition is insufficient. Active interaction selection will be tested only after this relational identification test succeeds. Task learning remains a later use case rather than part of E1.

A Implementation and evidence details

A.1 Staged learning protocol

The planned implementation follows the dependency structure of the hypothesis:

1. learn a persistent state and calibrate TP posterior, TP prior, and TP effect prediction;
2. set $R_D = 0$ and form provisional mechanisms from response families;
3. compare mechanism functions on unseen states, strengths, and combinations, then remove only functionally redundant laws;
4. enable relation residuals while freezing or strongly slowing mechanism functions;
5. jointly refine mechanisms and relations at lower learning rates while protecting isolated mechanism baselines;
6. use unresolved structural hypotheses to select new interactions;
7. only after world learning, infer task constraints from grouped demonstrations and realise proposed TPs through prior inversion.

An overcomplete bank is used so that the model need not know the true mechanism count. Sparsity encourages local explanations but is never accepted as evidence of identity by itself. A law is retained

when removing it causes a consistent held-out loss; two laws are merged only when one can replace the other across the relevant conditions without damaging previously explained data. ‘r’n